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PLA:

Pop:

$$\Sigma \begin{matrix} \nearrow n \\ \searrow p \end{matrix} \rightarrow \lambda_1 > \lambda_2 > \dots > \lambda_p > 0$$

$$y_j = a_j^T X; \quad 1 \leq j \leq p$$

$$a_j = e_j$$

~~Var(y_j)~~

$$\boxed{\text{Var}(y_j) = \lambda_j}$$

$$\sum e_j = \lambda_j e_j$$

$$y_j = e_j^T X = e_{j1} x_1 + e_{j2} x_2 + \dots + e_{jp} x_p$$

$\downarrow$
 $j^{\text{th}} \text{ PC}$
 $1 \leq j \leq p$

$\swarrow \quad \quad \quad \searrow$
loadings on $j^{\text{th}} \text{ PC}$

$$\begin{aligned}
 \text{Total variation of population} &= \text{Tr}(\Sigma) \\
 &= \text{Tr}(P \Lambda P^T) \\
 &= \text{Tr}(\Lambda P P^T) \\
 &= \text{Tr}(\Lambda) = \sum_{i=1}^p \lambda_i
 \end{aligned}$$

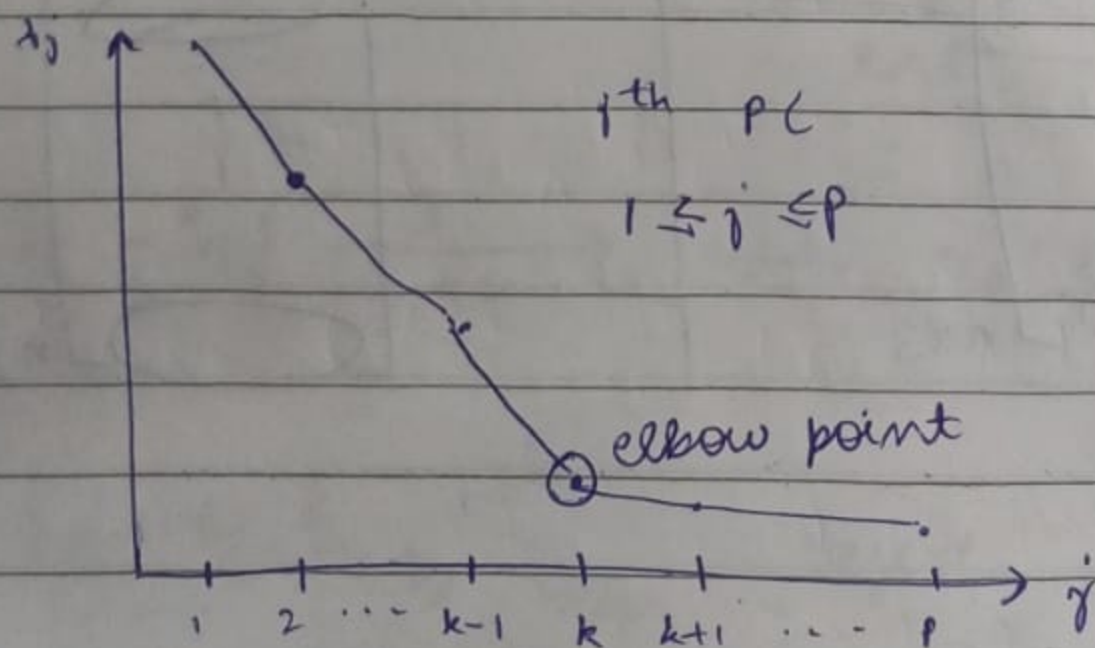
The j^{th} PC captures λ_j of total variation

$$\lambda_j / \left(\sum_{i=1}^p \lambda_i \right) \quad \left(\because \text{cov}(y_i, y_k) = 0 \quad i \neq k \right)$$

From the first k PC, the variation captured is

$$\frac{\lambda_1 + \dots + \lambda_k}{\left(\sum_{i=1}^p \lambda_i \right)} \times 100\%$$

we find smallest integer k at which this % is large enough to capture population total variation



discrete grid points

e.g. $\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$\text{cov}(X_3, X_1) = 0$
 $\text{cov}(X_3, X_2) = 0$
 Hence we expect X_3 to be one

$$\left. \begin{array}{l} \lambda_1 = 5.83 \\ \lambda_2 = 2 \\ \lambda_3 = 0.17 \end{array} \right\} \frac{\lambda_1 + \lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \times 100\% = 97.9\%$$

$$\lambda_1 \rightarrow e_1 = (0.383, -0.924, 0)^T$$

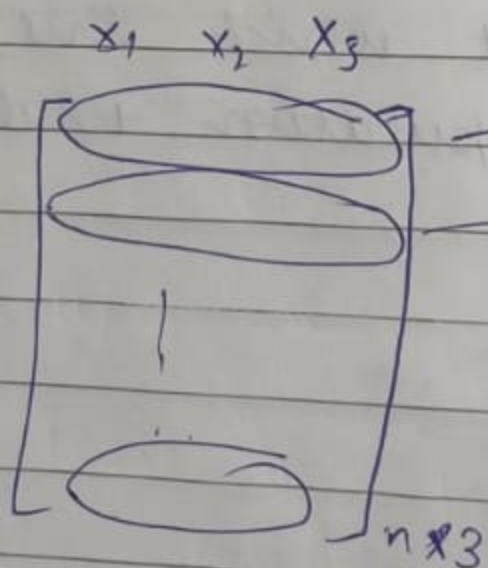
$$\lambda_2 \rightarrow e_2 = (0, 0, 1)^T$$

$$\lambda_3 \rightarrow e_3 = (0.924, 0.383, 0)^T$$

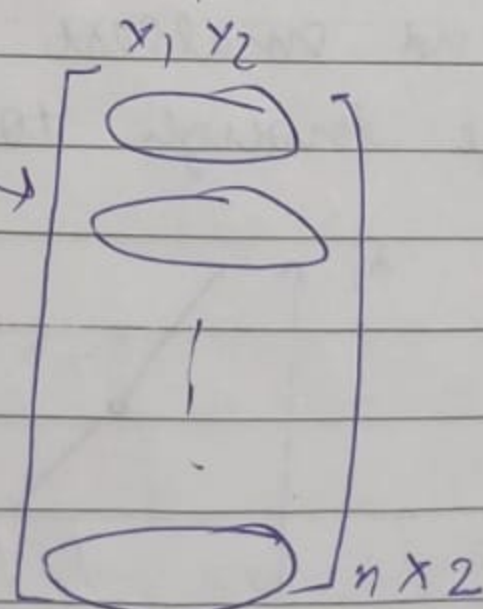
$$PC = \begin{cases} Y_1 = e_1^T X = 0.383 X_1 - 0.924 X_2 \\ Y_2 = e_2^T X = X_3 \end{cases}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} e_1^T \\ e_2^T \end{bmatrix} X = U^T X ; U = \begin{bmatrix} | & | \\ e_1 & e_2 \\ | & | \end{bmatrix}$$

Sample data matrix



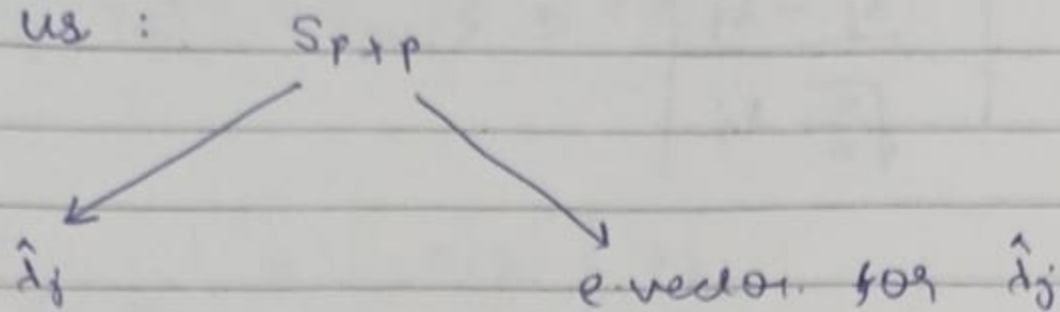
PC data matrix



Practically,

we will have data $n \times p$ given to us.

we have sample variance-covariance matrix given to us:



$$\hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_p > 0$$

$$S \hat{e}_j = \hat{\lambda}_j \hat{e}_j$$

$$\hat{Y}_j = \hat{e}_j^T X$$

choose first k of them:

$$\begin{array}{ccc} \hat{Y} & = & \hat{U}^T X \rightarrow p \times 1 \\ \downarrow & & \downarrow \\ k+1 & & k \times p \end{array}$$

$\Sigma \rightarrow$ not known

$\rightarrow \lambda$'s and e 's are not known

Through sample, we are estimating

$\hat{\lambda}, \hat{e} \rightarrow$ estimates for λ & e

Suppose population is $N_p(\mu, \Sigma)$

$$\downarrow \\ P \Lambda P^T$$

$$\sqrt{n} (\hat{\lambda}_j - \lambda_j) \xrightarrow[n \rightarrow \infty]{d} N_p(0, 2\lambda_j^2), \quad 1 \leq j \leq p$$

$$\left| \frac{\hat{\lambda}_j - \lambda_j}{\sqrt{\frac{2\lambda_j^2}{n}}} \right| > z_{\alpha/2}$$

reject $H_0: \lambda_j = \hat{\lambda}_j$ at α level of significance

$$H_1: \lambda_j \neq \hat{\lambda}_j$$

CI of for λ_j

$$\left| \frac{\hat{\lambda}_j - \lambda_j}{\sqrt{\frac{2}{n}} \lambda_j} \right| \leq z_{\alpha/2}$$

$$\left(1 - z_{\alpha/2} \sqrt{\frac{2}{n}}\right) \lambda_j \leq \hat{\lambda}_j \leq \left(1 + z_{\alpha/2} \sqrt{\frac{2}{n}}\right) \lambda_j$$

$$\Rightarrow \lambda_j \leq \frac{\hat{\lambda}_j}{1 - z_{\alpha/2} \sqrt{\frac{2}{n}}} \quad , \quad \lambda_j \geq \frac{\hat{\lambda}_j}{1 + z_{\alpha/2} \sqrt{\frac{2}{n}}}$$

$$\lambda_j \xrightarrow[\text{when } n \rightarrow \infty]{\text{CI}} \left[\frac{\hat{\lambda}_j}{1 - z_{\alpha/2} \sqrt{\frac{2}{n}}} , \frac{\hat{\lambda}_j}{1 + z_{\alpha/2} \sqrt{\frac{2}{n}}} \right]$$

$$n \rightarrow \infty \quad \sqrt{n} (\hat{e}_j - e_j) \xrightarrow{d} N_p(0, E_j)$$

$$E_k = \sum_{\substack{j=1 \\ j \neq k}}^p \frac{\lambda_j}{(\lambda_j - \lambda_k)^2} P_j e_j^T$$

Sample data matrix

If there are units attached to data then first correct data with standard data

$$\frac{x_i - \bar{x}_i}{\sqrt{s_{ii}}} \quad ; \quad 1 \leq i \leq p \quad \text{for every column}$$

then apply PCA with this standard data.

$$y_j = a_j^T X = e_j^T X$$

$$\text{var}(y_j) = \lambda_j$$

$$\text{cov}(y_j, y_k) = 0 \quad ; \quad j \neq k$$

Result: $\rho_{y_j, x_k} = e_{jk} \sqrt{\frac{\lambda_j}{\sigma_{kk}}}$

$$y_j = e_{j1}x_1 + \dots + \underbrace{e_{jk}x_k}_{\text{loading}} + \dots + e_{jp}x_p$$

Pf: ~~write~~ ~~write~~ write

$$x_k = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}_{1 \times p}^{X_{p \times 1}} = e^T X$$

k^{th} pos. $\swarrow$ $\searrow$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}$$

$$\begin{aligned} \text{cov}(y_j, x_k) &= \text{cov}(e_j^T X, e^T X) \\ &= e_j^T \Sigma e = e^T \Sigma e_j \quad (\Sigma^T = \Sigma) \\ &= e^D \lambda_j e_j = \lambda_j e^T e_j \end{aligned}$$

$$= \lambda_j e_{jk}$$

$$\text{cor}(y_j, x_k) = \frac{\text{cov}(y_j, x_k)}{\sqrt{\text{var}(y_j)} \sqrt{\text{var}(x_k)}}$$

$$= \frac{\lambda_j e_{jk}}{\sqrt{\lambda_j} \sqrt{\sigma_{kk}}} = e_{jk} \sqrt{\frac{\lambda_j}{\sigma_{kk}}}$$

e.g. - $\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$p = 2$

$\lambda_1 = 5.83 ; \lambda_2 = 2.0 ; \lambda_3 = 0.17$
 λ_3 ignored

$e_1 = \begin{bmatrix} 0.383 \\ -0.924 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$y_1 = 0.383 x_1 - 0.924 x_2$
 $y_2 = x_3$

$\text{cor}(y_1, x_1) = e_{11} \cdot \sqrt{\frac{\lambda_1}{\sigma_{11}}} = 0.383 \sqrt{\frac{5.83}{1}}$

$= 0.925 \rightarrow$ very high even tho loading was low

$\text{cor}(y_1, x_2) = -0.924 \sqrt{\frac{2}{5}} = -0.998$

Note: Standardized sample data

sample $\leftarrow \frac{x_i - \bar{x}_i}{\sqrt{\sigma_{ii}}}; 1 \leq i \leq p$

$\begin{pmatrix} \text{standard} \end{pmatrix}_{n \times p}$

var-cov of this data: $R = \text{correlation matrix}$

$$R = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \rightarrow \in (-1, 1)$$

↓
symm. pd.

$$\hat{p} = \text{corr}(y_k, \hat{y}_i) \rightarrow \text{standardized sample}$$

$$\hat{p} = \hat{e}_{ik} \sqrt{\hat{\lambda}_i}$$

Factor model (population)

$$x_1 = \mu_1 + l_{11} F_1 + \dots + l_{1m} F_m + e_1$$

$$x_2 = \mu_2 + l_{21} F_1 + \dots + l_{2m} F_m + e_2$$

$$x_p = \mu_p + l_{p1} F_1 + \dots + l_{pm} F_m + e_p$$

$$\hat{X} = \mu + \overset{p \times m}{L} \overset{m \times 1}{F} + \underset{\sim}{e}$$

loading matrix

→ unexplainable factor specific to feature

$F_1, \dots, F_m$

m basis vectors

→ explainable but unobservable
→ influencing the entire or all the features